

TIRSIT BERIHU

AI/ML ENGINEER

TirsitTeshome



Tirsit's Portfolio



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Tirsit-Berihu



PERSONAL STATEMENT

With a keen eye for aesthetics and a drive to combine functionality with art, Tirsit is deeply inspired by the power of visuals and how they influence perception. Her love for photography showed her that, like art, technology can communicate through images. As a software engineer, she's driven to bring that same visual storytelling to life. She's excited to contribute to an industry that allows her to seamlessly blend aesthetics with robust functionality.

EDUCATION

AkiraChix

codeHive - Diploma in Information Technology
February 2025 - present

SKILLS

Languages

Python
SQL

Libraries & Frameworks

Data acquisition & cleaning: Pandas, spaCy, BeautifulSoup, Tika, WordNinja, PDFPlumber
Embeddings: SentenceTransformers (all-MiniLM-L6-v2), Text-embedding 004, Huggingface Transformer
Databases: PostgreSQL, ChromaDB
Vector search: PGVector(PostgreSQL extension)
AI/ Machine learning: scikit-learn, ADK, NLP, prompt engineering, Google ADK, Retrieval-Augmented Generation (RAG)
Deep learning: TensorFlow/Keras, PyTorch,
Visualization: Matplotlib, Seaborn

PROJECTS

- Movie recommendation agent- [link](#)
- Breast Cancer image classification - [link](#)
- Zeno AI-[link](#)
- House price prediction- [link](#)

REFERENCE

Linda Kamau

Founder and Executive Director
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Synthia Hunter Achieng

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EXPERIENCES

Zeno AI

Zeno is an AI assistant for economists in Kenya. It takes data on maize, coffee, and tea and turns it into clear insights, such as identifying the impact of tariffs on Kenya's agricultural trade competitiveness, allowing economists to focus on what they do best, making smarter decisions for Kenya's agricultural trade.

- Collected agricultural data on maize, coffee, and tea from various file formats using PDFPlumber to extract tables, BeautifulSoup4 for web scraping, and Pandas to organize the data, ensuring clean and well-structured information for analysis.
- Split datasets into 600-character chunks with a 100-character overlap to keep context between segments, then generated 768-dimensional embeddings, which helped improve the quality of text representation.
- Used the Text-embedding-004 model to create embeddings that captured richer context compared to the Hugging Face (BAAI/bge-m3) model, leading to better understanding of the data.
- Stored embeddings in PGVector within PostgreSQL to keep all vector data in one place, making integration easier and simplifying deployment for production.
- Applied hybrid Chain-of-Thought and Zero-Shot prompting methods to support detailed multi-step economic reasoning, enhancing the model's ability to analyze trade scenarios.
- Built and containerized an interactive agent with Google ADK for natural language Q&A on key crops, then deployed it on Google Cloud for easy use and management.

House Price Prediction

Built a system that predicts home sale prices by analyzing important property features and understanding how they impact prices, leading to more accurate predictions.

- Performed feature engineering (one-hot encoding, log-scaling of skewed variables) and missing-value imputation with Pandas/NumPy to turn raw data into features that improve machine learning model performance.
- Compared several models such as Linear Regression, Lasso, SVR, RandomForest, XGBoost using Root Mean Squared Error(RMSE) and selected XGBoost as the final estimator because it achieved the lowest error with an RMSE of approximately 0.127. It reduced prediction error by about 18% compared to Linear Regression and 12% compared to RandomForest, while SVR showed the weakest generalization and underperformed on large datasets.
- Visualized feature contributions and feature importance using Matplotlib and Seaborn to better understand and interpret how each feature influences the model's predictions.

Breast Cancer Image Classification

Built a deep-learning model to predict the category of breast tumors, benign, malignant, or normal, based on ultrasound images.

- Applied augmentation (rotation, zoom, flips, shears) via ImageDataGenerator to expand the training dataset and increase the diversity of image examples.
- Experimented with custom CNNs and MobileNetV2 transfer learning; MobileNetV2 achieved ~93 % validation accuracy, outperforming a scratch-built CNN by 11 percentage points and reducing overfitting.
- Designed a testing pipeline to preprocess and classify unseen scans, mapping probability scores to clinical labels.

Movie Recommendation Agent

Built a movie recommendation agent designed to help users find films that suit their personal preferences.

- Loaded and cleaned 1000-movie IMDB dataset using Pandas/NumPy, enhancing data quality for recommendation pipelines.
- Used Sentence Transformers (all-MiniLM-L6-v2) for text embeddings and ChromaDB for efficient vector search, enabling fast semantic queries.
- Integrated Google ADK Agent to orchestrate data loading, retrieve top-5 results, and deliver witty, context-aware explanations.
- Developed a custom explain_choice module with randomized templates to keep responses engaging when the Gemini API is unavailable.
- Established a persistent vector database storing embeddings, achieving one-second retrieval latency for similarity searches across movies.